

Q1: Gambler's Ruin with Draws

[15 marks]

We have $p = q$, $s = 1 - 2p$, and $0 < p < 1/2$.

(a) Ruin probability

[4 marks]

Let r_i denote the probability that Alice is eventually ruined given that her current fortune is i . Conditioning on the outcome of the next round,

$$\begin{aligned} r_i &= \Pr(\text{ruin} \mid X_0 = i) \\ &= \Pr(\text{ruin} \mid X_1 = i + 1, X_0 = i) \Pr(X_1 = i + 1 \mid X_0 = i) \\ &\quad + \Pr(\text{ruin} \mid X_1 = i - 1, X_0 = i) \Pr(X_1 = i - 1 \mid X_0 = i) \\ &\quad + \Pr(\text{ruin} \mid X_1 = i, X_0 = i) \Pr(X_1 = i \mid X_0 = i) \\ &= pr_{i+1} + pr_{i-1} + sr_i, \end{aligned}$$

where the last equality follows since, conditional on Alice's fortune after the first round, the probability of eventual ruin depends only on her new fortune. Hence $pr_{i+1} - (1 - s)r_i + pr_{i-1} = 0$. The boundary conditions are $r_0 = 1$ and $r_M = 0$.

Since $1 - s = 2p$, dividing by p gives $r_{i+1} - 2r_i + r_{i-1} = 0$. Assume $r_i = \lambda^i$. The characteristic equation is

$$\lambda^2 - 2\lambda + 1 = (\lambda - 1)^2 = 0.$$

Thus $\lambda = 1$ is a repeated root and $r_i = A + Bi$. Using $r_0 = 1$ gives $A = 1$, while $r_M = 0$ gives $B = -1/M$. Therefore

$$r_i = 1 - \frac{i}{M} = \frac{M - i}{M}.$$

(b) Embedded random walk

[2 marks]

Money is exchanged in a round with probability $p + q = 2p$. Conditional on an exchange,

$$\Pr(\text{Alice wins} \mid \text{exchange}) = \frac{p}{2p} = \frac{1}{2}, \quad \Pr(\text{Alice loses} \mid \text{exchange}) = \frac{p}{2p} = \frac{1}{2}.$$

Hence, for $1 \leq i \leq M - 1$,

$$\Pr(Z_{k+1} = i + 1 \mid Z_k = i) = \Pr(Z_{k+1} = i - 1 \mid Z_k = i) = \frac{1}{2}.$$

Moreover, since the outcomes of successive original rounds are independent, the outcomes of successive exchange rounds are also independent. Therefore, the increments $Z_{k+1} - Z_k$ are i.i.d., with

$$\Pr(Z_{k+1} - Z_k = 1) = \Pr(Z_{k+1} - Z_k = -1) = \frac{1}{2},$$

until the process reaches 0 or M .

Although $\{Z_k\}$ records only the rounds in which money is exchanged, the original game terminates as soon as Alice's fortune reaches 0 or M . Consequently, once Z_k reaches either boundary, no further round, and hence no further exchange round, is played. Thus, 0 and M are absorbing states for $\{Z_k\}$.

Therefore, $\{Z_k\}$ is a simple symmetric random walk on $\{0, 1, \dots, M\}$, absorbed at 0 and M .

(c) Expected number of money-exchange rounds**[3 marks]**

At the end of the embedded random walk, $Z_N \in \{0, M\}$. From part (a),

$$P(Z_N = 0) = r_a = \frac{M-a}{M}, \quad P(Z_N = M) = 1 - r_a = \frac{a}{M}.$$

Therefore,

$$\mathbb{E}[Z_N^2] = 0^2 \frac{M-a}{M} + M^2 \frac{a}{M} = aM.$$

Using the given identity $\mathbb{E}[N] = \mathbb{E}[Z_N^2] - a^2$, we obtain

$$\mathbb{E}[N] = aM - a^2 = \boxed{a(M-a)}.$$

(d) Waiting time between successive exchanges**[3 marks]**

In each original round, money is exchanged with probability $2p$. Therefore

$$\boxed{G_k \sim \text{Geom}(2p)}$$

on $\{1, 2, \dots\}$, with

$$P(G_k = j) = (1 - 2p)^{j-1}(2p) = s^{j-1}(2p), \quad j \geq 1.$$

Hence $\mathbb{E}[G_k] = 1/(2p)$. Since the original rounds are independent and successive waiting periods concern disjoint blocks of rounds, $G_1, G_2, \dots$ are independent and identically distributed. Thus

$$\boxed{G_1, G_2, \dots \text{ are i.i.d. } \text{Geom}(2p), \quad \mathbb{E}[G_k] = \frac{1}{2p}.$$

(e) Expected total duration**[3 marks]**

If N money exchanges are needed for absorption, then

$$\boxed{T = \sum_{k=1}^N G_k.}$$

Conditioning on $N = n$,

$$\mathbb{E}[T \mid N = n] = \mathbb{E}\left[\sum_{k=1}^n G_k\right] = \sum_{k=1}^n \mathbb{E}[G_k] = \frac{n}{2p}.$$

Therefore $\mathbb{E}[T \mid N] = N/(2p)$ and, by the nested expectation property,

$$\mathbb{E}[T] = \mathbb{E}[\mathbb{E}[T \mid N]] = \frac{\mathbb{E}[N]}{2p} = \boxed{\frac{a(M-a)}{2p}}.$$

Equivalently, Wald's identity after justifying the required stopping assumptions can be used.

Q2: Conditional Arrival Time**[2 marks]**

For $0 \leq x \leq t$, the event $\{S_1 \leq x, N_t = 1\}$ means that there is exactly one arrival in $(0, x]$ and no arrivals in $(x, t]$.

$$\Pr(S_1 \leq x, N_t = 1) = \Pr(N_x = 1, N_t - N_x = 0).$$

By independent increments,

$$\Pr(N_x = 1, N_t - N_x = 0) = \Pr(N_x = 1) \Pr(N_t - N_x = 0).$$

Also, by stationary increments,

$$N_t - N_x \stackrel{d}{=} N_{t-x},$$

and hence

$$\Pr(N_t - N_x = 0) = \Pr(N_{t-x} = 0) = e^{-\lambda(t-x)}.$$

Therefore,

$$\Pr(S_1 \leq x, N_t = 1) = (\lambda x) e^{-\lambda x} e^{-\lambda(t-x)} = \lambda x e^{-\lambda t}.$$

Also, $P(N_t = 1) = \lambda t e^{-\lambda t}$. Hence

$$P(S_1 \leq x \mid N_t = 1) = \frac{\lambda x e^{-\lambda t}}{\lambda t e^{-\lambda t}} = \left\lfloor \frac{x}{t} \right\rfloor.$$

Therefore

$$S_1 \mid \{N_t = 1\} \sim \text{Unif}(0, t].$$

Q3: Renewal Stopping Time

[3 marks]

Let $S_n = \sum_{i=1}^n X_i$. Since

$$N_k = \min\{n \geq 1 : S_n > k\},$$

we have $S_{N_k-1} \leq k < S_{N_k}$. Since $0 \leq X_i \leq 1$,

$$k < S_{N_k} \leq k + 1.$$

Taking expectations gives $k < \mathbb{E}[S_{N_k}] \leq k + 1$. Since $\mathbb{E}[X_1] = 1/2$, Wald's identity gives

$$\mathbb{E}[S_{N_k}] = \mathbb{E}[N_k] \mathbb{E}[X_1] = \frac{1}{2} \mathbb{E}[N_k].$$

Thus

$$2 < \frac{\mathbb{E}[N_k]}{k} \leq 2 + \frac{2}{k}.$$

By the squeeze theorem,

$$\lim_{k \rightarrow \infty} \frac{\mathbb{E}[N_k]}{k} = 2.$$

Q4: Ergodicity

[5 marks]

(a) Mean ergodicity

[3 marks]

Define

$$\bar{X}_N = \frac{1}{N} \sum_{n=0}^{N-1} X[n].$$

Since the process is WSS, $\mathbb{E}[X[n]] = \mu$ for every n , and hence

$$\mathbb{E}[\bar{X}_N] = \frac{1}{N} \sum_{n=0}^{N-1} \mathbb{E}[X[n]] = \mu.$$

Therefore

$$\mathbb{E}[(\bar{X}_N - \mu)^2] = \text{Var}(\bar{X}_N),$$

so

$$\bar{X}_N \xrightarrow{\text{m.s.}} \mu \iff \lim_{N \rightarrow \infty} \text{Var}(\bar{X}_N) = 0.$$

Now,

$$\begin{aligned}\text{Var}(\bar{X}_N) &= \text{Var}\left(\frac{1}{N} \sum_{n=0}^{N-1} X[n]\right) \\ &= \frac{1}{N^2} \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} \text{Cov}(X[n], X[m]).\end{aligned}$$

Since $\{X[n]\}$ is WSS, $\text{Cov}(X[n], X[m]) = C_X[n - m]$. Hence,

$$\text{Var}(\bar{X}_N) = \frac{1}{N^2} \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} C_X[n - m].$$

For a fixed lag $k = n - m$, there are $N - |k|$ pairs (n, m) such that $n - m = k$. Therefore,

$$\begin{aligned}\text{Var}(\bar{X}_N) &= \frac{1}{N^2} \sum_{k=-(N-1)}^{N-1} (N - |k|) C_X[k] \\ &= \frac{1}{N} \sum_{k=-(N-1)}^{N-1} \left(1 - \frac{|k|}{N}\right) C_X[k].\end{aligned}$$

Separating the $k = 0$ term,

$$\begin{aligned}\text{Var}(\bar{X}_N) &= \frac{C_X[0]}{N} + \frac{1}{N} \sum_{k=1}^{N-1} \left(1 - \frac{k}{N}\right) C_X[k] \\ &\quad + \frac{1}{N} \sum_{k=1}^{N-1} \left(1 - \frac{k}{N}\right) C_X[-k].\end{aligned}$$

For a WSS process, the covariance function is even, i.e., $C_X[-k] = C_X[k]$. Hence,

$$\boxed{\text{Var}(\bar{X}_N) = \frac{C_X[0]}{N} + \frac{2}{N} \sum_{k=1}^{N-1} \left(1 - \frac{k}{N}\right) C_X[k].}$$

Since $C_X[0] = \text{Var}(X[n]) < \infty$ for a WSS process,

$$\frac{C_X[0]}{N} \rightarrow 0.$$

Therefore, $\{X[n]\}$ is ergodic in mean if

$$\boxed{\lim_{N \rightarrow \infty} \frac{2}{N} \sum_{k=1}^{N-1} \left(1 - \frac{k}{N}\right) C_X[k] = 0.}$$

(b) Autocorrelation ergodicity

[1 mark]

For fixed k ,

$$\hat{R}_N[k] = \frac{1}{N} \sum_{n=0}^{N-1} X[n]X[n+k].$$

WSS implies $\mathbb{E}[\hat{R}_N[k]] = R_X[k]$. However, mean-square convergence also requires $\text{Var}(\hat{R}_N[k]) \rightarrow 0$. Expanding this variance produces terms of the form

$$\mathbb{E}[X[n]X[n+k]X[m]X[m+k]],$$

which are fourth-order moments. WSS specifies only first- and second-order statistics and therefore does not determine these quantities. Hence

WSS alone is not sufficient; fourth-order statistical information is required.

(c) Gaussian case

[1 mark]

For a Gaussian process, fourth-order moments are completely determined by first- and second-order moments. Hence, the WSS property is enough to analyze ergodicity in autocorrelation for a Gaussian process.

Q5: Poisson Process from Exponential Interarrival Times

[5 marks]

Let $S_n = \sum_{i=1}^n X_i$, where X_i are i.i.d. $\text{Exp}(\lambda)$, and let $N(t) = \max\{n : S_n \leq t\}$.

(a) Residual waiting time

[2 marks]

Let $R_t = S_{N(t)+1} - t$. Suppose $N(t) = n$ and $S_n = u \leq t$. Since there has been no arrival between u and t , we know $X_{n+1} > t - u$. For $x \geq 0$,

$$\begin{aligned} P(R_t > x \mid N(t) = n, S_n = u) &= P(X_{n+1} > t - u + x \mid X_{n+1} > t - u) \\ &= e^{-\lambda x}, \end{aligned}$$

where the last equality follows from the memoryless property of the exponential distribution. Since this expression does not depend on n or u , averaging over $N(t)$ and $S_{N(t)}$ gives

$$P(R_t > x) = e^{-\lambda x}, \quad x \geq 0.$$

Therefore

$$\boxed{R_t \sim \text{Exp}(\lambda)}.$$

(b) Restart property

[1 mark]

Part (a) shows that the waiting time from t to the next arrival is again $\text{Exp}(\lambda)$. After that arrival, the subsequent interarrival times are again independent $\text{Exp}(\lambda)$ random variables. Hence, viewed from time t , the sequence of future interarrival times has the same joint distribution as the original sequence beginning at time 0. Therefore the arrival process observed after time t has the same distribution as a fresh copy of the original process.

(c) Stationary increments

[2 marks]

Define $\tilde{N}(u) = N(t+u) - N(t)$ for $u \geq 0$. This counts the arrivals in $(t, t+u]$. By part (b), the process observed after time t has the same distribution as the original process starting at 0. Hence

$$\boxed{\{\tilde{N}(u) : u \geq 0\} \stackrel{d}{=} \{N(u) : u \geq 0\}}.$$

Equivalently,

$$\boxed{\{N(t+u) - N(t) : u \geq 0\} \stackrel{d}{=} \{N(u) : u \geq 0\}}.$$

In particular, for every $t, h \geq 0$,

$$\boxed{N(t+h) - N(t) \stackrel{d}{=} N(h)}.$$

Thus the distribution of an increment depends only on the interval length h , and $\{N(t)\}_{t \geq 0}$ has stationary increments.